

# Memory Snap

A Picture-Recall Cognitive Game for Elders, with a Caregiver Content & Monitoring Portal

Complete Implementation Plan & Technical Specification (Manual Content Model, No AI)

*Prepared for build-out with an AI app-building agent (e.g. Antigravity)*

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# 1. Executive Summary

Memory Snap is a web application designed to give elderly users a light, friendly cognitive exercise: an image is shown for a short viewing period, then a simple question is asked about what was just seen (e.g. "What color was the umbrella?", "How many people were in the photo?"). All content is fully caregiver-authored: the caregiver uploads each photo and writes the question(s) and answer choices that go with it directly. There is no AI-driven content generation anywhere in this version — every photo and every question the elder ever sees was deliberately added by a caregiver, and can be edited or removed by that caregiver at any time.

A separate caregiver portal lets a caregiver (family member, aide, or care-facility staff) create an elder's profile, upload images with their matching questions, manage that content library (add, edit, remove), configure session settings, and monitor accuracy, response time, and trends over time. The elder-side gameplay logic — show the image, hide it, ask the question, give gentle feedback — is unchanged from the original concept. This document gives an AI coding agent (Antigravity, or similar) everything needed to build it: architecture, data model, API contracts, screen specs, and a phased build order.

# 2. Goals & Success Metrics

- **Cognitive engagement:** give elders a short, repeatable visual-memory exercise (2–5 minutes per session).
- **Full caregiver control:** every photo and question is hand-authored by the caregiver — nothing is auto-generated, so content quality and relevance are entirely in the caregiver's hands.
- **Personalization:** caregivers use meaningful images (family, pets, home objects) paired with questions they know the elder can reason about.
- **Caregiver visibility:** surface accuracy, response time, and streaks so a caregiver can notice if performance changes over time.
- **Low friction for elders:** large text, minimal steps, no logins-within-logins, voice-friendly, forgiving of mis-taps.
- **Simple content management:** adding, editing, and removing photos/questions should take a caregiver well under a minute per item.

## Success metrics to track post-launch:

- % of started sessions completed without the elder abandoning mid-way.
- Average accuracy per elder over rolling 7/30-day windows.
- Number of photo+question pairs a caregiver maintains and how often they refresh the library.
- Time-on-task per question, as a proxy for cognitive load/fatigue.

# 3. Users & Roles

| Role           | Description                                     | Primary Needs                                                        |
|----------------|-------------------------------------------------|----------------------------------------------------------------------|
| Elder (Player) | The person playing the memory game.             | Extremely simple UI, large fonts/buttons, minimal typing, clear feed |
| Caregiver      | Family member, aide, or facility staff managing | Fast photo+question profiles, easy editing/removal, readable perform |

| Role                 | Description                                                                             | Primary Needs  |
|----------------------|-----------------------------------------------------------------------------------------|----------------|
| Admin (optional, v2) | Product/support admin overseeing multiple caregiver accounts (for facility deployments) | System health. |

## 4. Core Feature Set

### 4.1 Elder-Facing

- Simple "Start Game" screen with the elder's name/avatar.
- Image display step: image shown full-screen for a configurable duration (default 8–12 seconds), with a large countdown ring.
- Question step: image is hidden, one question is asked with 2–4 large tappable answer choices (multiple choice by default).
- Immediate, gentle feedback (checkmark / "Good try!") — never harsh or game-over messaging.
- Session summary screen: simple stars or a smiling-face scale, not raw percentages, at the end of each session.
- Optional read-aloud (text-to-speech) of questions and instructions.

### 4.2 Caregiver-Facing

- Caregiver account with one or more linked elder profiles.
- Add-a-card flow: upload one photo, then write its question, correct answer, and 1–3 wrong-answer choices — all in a single short form.
- Content library view: every photo the caregiver has added, each showing its attached question(s), with Edit and Delete controls.
- Edit any existing photo's question/choices at any time; delete a photo (and its questions) entirely, instantly removing it from the elder's rotation.
- Dashboard: accuracy over time, response-time trend, session frequency, streaks, per-question breakdown.
- Session configuration: difficulty (drives viewing duration and number of answer choices), number of questions per session.
- Downloadable/printable weekly summary (PDF or CSV export).

### 4.3 Shared / System

- No AI, no auto-generation, no image recognition anywhere in the pipeline — 100% of playable content is caregiver-authored.
- Each elder's question pool is simply the full set of photo+question cards the caregiver has added for that elder (or shared across their elders, if the caregiver manages more than one).
- Basic upload validation (file type/size) — no AI safety scanning needed since only the caregiver themselves is adding content.

## 5. Elder Game Flow (UX Walkthrough)

This flow is unchanged from the original concept:

### 1. Welcome

Elder taps their name/photo on a shared family tablet, or the app opens directly to their profile (single-elder device). No password required — a caregiver can set a 4-digit PIN or nothing at all.

### 2. Instructions (first time only)

One friendly sentence + a demo round: "Look at the picture, then answer a simple question about it."

### 3. Image Reveal

Full-screen image, large countdown ring around the edge. No buttons visible yet, so nothing to accidentally tap.

### 4. Question

Image fades out, question text appears large and centered, 2–4 big buttons below (or read aloud automatically). The question and choices shown are exactly what the caregiver typed in when the photo was added.

### 5. Feedback

Immediate colored highlight on the chosen answer + short encouraging text/sound. No score shown mid-game.

### 6. Repeat

3–8 rounds per session, drawn at random from that elder's caregiver-authored card pool, configurable by the caregiver.

### 7. Session End

A simple, warm summary (stars, a cheerful character, or "You did great today!") — never a harsh numeric score front-and-center.

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## 6. Caregiver Portal Flow (UX Walkthrough)

- **Sign up / Login** — standard email+password or Google/Apple sign-in.
  - **Add Elder Profile** — name, optional photo, birth year (for age-appropriate defaults), difficulty starting point.
  - **Add a Card (Photo + Question)** — upload a photo, then in the same form: type the question text, the correct answer, and 1–3 wrong-answer choices. Save adds it immediately to that elder's live pool.
  - **Manage Library** — browse all cards for an elder as a grid; each card shows a thumbnail + its question; Edit opens the same form pre-filled; Delete removes the card (with a confirmation step) and it instantly stops appearing in new sessions.
  - **Configure Sessions** — set reveal duration, number of rounds, and difficulty default for the elder.
  - **Dashboard** — charts for accuracy trend, response time trend, streak calendar, and a plain-language weekly note (e.g. "Accuracy dipped slightly this week compared to last").
  - **Export / Share** — download a PDF/CSV report, useful for sharing with a doctor or family member.
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## 7. Accessibility & Elder-Friendly Design Guidelines

| Guideline                        | Implementation Detail                                                                                          |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| Large, high-contrast text        | Minimum 20–24px body text on the elder-facing views; WCAG AA contrast ratio (4.5:1) minimum, prefer AAA (7:1). |
| Big tap targets                  | Buttons at least 64x64px with generous spacing to prevent mis-taps from tremor or reduced dexterity.           |
| No time pressure on answers      | Only the image-viewing step is timed; the question step waits for the elder's response.                        |
| Audio support                    | Built-in text-to-speech for all instructions and questions; adjustable playback speed.                         |
| Minimal navigation depth         | Elder view has effectively one screen at a time, no nested menus, no back-button confusion.                    |
| Consistent, calm visual language | Soft color palette, no flashing elements, no aggressive game-over or failure states.                           |
| Forgiving interaction            | Mis-taps are easy to correct; no penalty for taking time to answer.                                            |
| Device flexibility               | Responsive layout for tablet (primary target), also usable on desktop browser and phone.                       |

## 8. System Architecture

The system is a standard three-tier web app: a frontend SPA (two experiences — Elder Mode and Caregiver Mode — sharing one codebase), a backend API, and a database. There is no external AI/vision service in this version — the pipeline is entirely CRUD: the caregiver creates a photo+question record, and the elder's session simply reads from that record.

### High-level components:

- **Frontend (SPA):** React app with two route groups — `/play/*` (Elder Mode, kiosk-style) and `/caregiver/*` (dashboard, auth-gated).
- **Backend API:** REST service handling auth, profiles, image storage references, card (photo+question) CRUD, session logic, and scoring.
- **Database:** relational store (Postgres) for structured data — users, elders, cards (photo + question + choices), sessions, answers.
- **Object storage:** S3-compatible bucket for caregiver-uploaded images.
- **Text-to-Speech Service:** browser-native Web Speech API for v1 (no backend cost); optional higher-quality TTS API in v2.
- **Notification Service:** email (and optional SMS/push in v2) for weekly caregiver summaries and streak alerts.

### Request flow — adding a new card:

1) Caregiver selects a photo and fills in the question + correct answer + wrong-answer choices in one form → 2) Frontend uploads the photo to object storage via a signed URL → 3) Frontend submits the card (photo reference + question text + choices) to the backend → 4) Backend validates and stores the card as immediately **active** → 5) The card is now eligible to appear in that elder's next session — no review or approval step, since the caregiver is the sole author.

### Request flow — removing a card:

1) Caregiver taps Delete on a card in the Library view → 2) Confirmation prompt → 3) Backend marks the card deleted (soft delete recommended, see Section 15) and it is excluded from all future session draws → 4) Past session\_answers referencing that card are kept for historical reporting, unaffected by the deletion.

## 9. Recommended Tech Stack

| Recommendation                                                                  | Notes                                                                           |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| JavaScript, Vite, Tailwind CSS                                                  | Two theme presets: "Elder" (large/simple) and "Caregiver" (dense/detailed)      |
| Redis (server cache) + lightweight local state                                  | Keeps session state simple during live gameplay.                                |
| TypeScript (Express or Fastify) or Python (FastAPI)                             | Either is fine — no AI/ML dependency to steer the choice, so pick what you like |
| A managed database (e.g. Supabase, Neon, RDS)                                   | Managed instance (e.g. Supabase, Neon, RDS) to reduce ops overhead              |
| S3 or Cloudflare R2 (image storage)                                             | Store originals + a resized display copy per image.                             |
| Auth (Google) via a managed auth provider (e.g. Supabase Auth, Auth0, Clerk)    | Caregiver-only; elder profiles are sub-records, not separate logins.            |
| OpenAI API (browser) for v1                                                     | No backend cost; quality is adequate for instructions/questions.                |
| Vercel (frontend) + Render/Fly.io/Railway (backend) or a single full-stack host | Pick whichever the build agent supports most directly.                          |
| Analytics (PostHog) + error tracking (Sentry)                                   | Helps catch elder-side friction (drop-off points) after launch.                 |

## 10. Data Model / Database Schema

Simplified relative to an AI-driven version: photo and question are combined into one caregiver-authored cards record, with no review/approval or AI-provenance fields needed.

```
caregivers
  id          uuid PK
  email       text unique
  password_hash text
  full_name   text
  created_at  timestampz

elders
  id          uuid PK
  caregiver_id uuid FK -> caregivers.id
  display_name text
  birth_year  int null
  avatar_url  text null
  pin_code    text null          -- optional, 4-digit
  difficulty_level text default 'easy' -- easy | medium | hard
  created_at  timestampz

cards
  id          uuid PK
  elder_id    uuid FK -> elders.id  -- card belongs to one elder's pool
                                           -- (or make nullable + a join table if a car
egiver
                                           -- wants to share one card across multiple
elders)
  caregiver_id uuid FK -> caregivers.id -- author, for auditing/ownership
  photo_url    text
  thumbnail_url text
  question_prompt text              -- "What color was the umbrella?"
  correct_choice text
  wrong_choices jsonb              -- ["Blue","Green","Yellow"]
  is_active    boolean default true  -- soft-delete flag, see Section 15
  created_at   timestampz
  updated_at   timestampz

sessions
  id          uuid PK
  elder_id    uuid FK -> elders.id
  started_at  timestampz
  completed_at timestampz null
  round_count int

session_answers
  id          uuid PK
  session_id  uuid FK -> sessions.id
  card_id     uuid FK -> cards.id
  selected_choice text
  is_correct  boolean
  response_time_ms int
  answered_at timestampz
```

Notes: **cards.is\_active** implements a soft delete — set to `false` instead of hard-deleting the row, so historical `session_answers` still resolve to a real card for reporting purposes even after a caregiver removes it from rotation. Only `is_active = true` cards are eligible to be drawn into a new session.

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## 11. API Endpoint Specification

| Method & Path                                   | Purpose                                                                                   |
|-------------------------------------------------|-------------------------------------------------------------------------------------------|
| POST /auth/signup, /auth/login                  | Caregiver account creation and login.                                                     |
| GET /elders                                     | List elder profiles for the logged-in caregiver.                                          |
| POST /elders                                    | Create a new elder profile.                                                               |
| PATCH /elders/:id                               | Update difficulty, PIN, session config, etc.                                              |
| POST /cards/upload-url                          | Request a signed URL for direct-to-storage photo upload.                                  |
| POST /cards                                     | Create a card: photo reference + question prompt + correct answer + wrong choices.        |
| GET /elders/:id/cards                           | List all cards (active and inactive) for an elder, for the Library view.                  |
| PATCH /cards/:id                                | Edit a card's photo, question, or choices.                                                |
| DELETE /cards/:id                               | Soft-delete a card (sets <code>is_active = false</code> ); excluded from future sessions. |
| POST /cards/:id/restore                         | (Optional) Reactivate a previously removed card.                                          |
| GET /play/:elderId/session                      | Start a new game session; server returns a random ordered set of active cards.            |
| POST /play/:elderId/session/:sessionId/answer   | Submit an answer for the current round (choice + response time).                          |
| POST /play/:elderId/session/:sessionId/complete | Mark session finished; triggers summary calculation.                                      |
| GET /elders/:id/stats                           | Aggregated accuracy, response-time trend, streaks for the dashboard.                      |
| GET /elders/:id/report.pdf                      | Generate a downloadable caregiver report.                                                 |

## 12. Manual Question Authoring & Content Management

There is no automated content generation. Every card is created by the caregiver filling out one short form. Keeping that form fast and clear is the most important UX detail in the whole caregiver portal, since it's the one repeated task that populates the entire game.

### "Add a Card" form fields:

- **Photo** — upload from device or drag-and-drop (required).
- **Question** — free-text prompt, e.g. "What was on the table?" (required).
- **Correct answer** — free-text, e.g. "A vase of flowers" (required).
- **Wrong-answer choices** — 1 to 3 free-text distractors the caregiver writes themselves, e.g. "A lamp", "A bowl of fruit" (at least 1 required; more choices = harder).

- **Save** — the card becomes active immediately; no review step, no waiting.

**Suggested lightweight guidance shown in the form (not enforced by validation, just helper text):**

- Keep the question about one clearly visible detail (a color, a count, a named object).
- Make wrong-answer choices plausible but clearly different from the correct answer.
- Avoid trick questions or anything requiring outside knowledge the elder wouldn't have from the photo alone.

**Example record as the caregiver enters it:**

|                 |                                |
|-----------------|--------------------------------|
| Photo:          | beach_photo_2024.jpg           |
| Question:       | "What color was the umbrella?" |
| Correct answer: | "Red"                          |
| Wrong choices:  | "Blue", "Green", "Yellow"      |

### 13. Difficulty Levels & Session Composition

Since questions are hand-written, "difficulty" here mainly controls presentation — viewing time and how many choices are shown — rather than the nature of the question itself (that's under the caregiver's own judgment when they write it).

| Level  | Viewing Time  | Choices Shown                 | Caregiver Guidance                                                       |
|--------|---------------|-------------------------------|--------------------------------------------------------------------------|
| Easy   | 12–15 seconds | 2 choices (correct + 1 wrong) | Write 1 wrong-answer choice; keep the question to one obvious detail.    |
| Medium | 8–10 seconds  | 3 choices (correct + 2 wrong) | Write 2 wrong-answer choices; can reference counts or secondary objects. |
| Hard   | 5–6 seconds   | 4 choices (correct + 3 wrong) | Write 3 wrong-answer choices; can reference less prominent details.      |

If a card has fewer wrong choices than the elder's current difficulty calls for, the session simply shows however many the caregiver provided — the form never blocks saving a card for having only 1 wrong choice.

### 14. Scoring, Progress Tracking & Reports

- **Per-session:** % correct, average response time, shown to the elder only as a warm qualitative summary (stars/smileys), never a bare percentage.
- **Per-elder trend (caregiver view):** line chart of accuracy over the last 30 days, response-time trend, session frequency/streak calendar.
- **Per-card performance:** caregiver can see which specific cards the elder consistently gets right or wrong — useful for deciding what to edit or remove.
- **Weekly digest:** auto-generated plain-language note (e.g. "5 sessions this week, accuracy steady around 78%") emailed to the caregiver.
- **Exportable report:** PDF/CSV suitable for sharing with a physician or care team.

### 15. Image & Question Lifecycle (Add, Edit, Remove)

## 15.1 Adding

- Caregiver uploads a photo and fills in the question form (Section 12) in one step.
- Card is active immediately — appears in the pool for the next randomly drawn session.

## 15.2 Editing

- Caregiver opens any card from the Library view; the same form appears pre-filled.
- Caregiver can replace the photo, change the question text, correct answer, or wrong choices, then save.
- Edits apply going forward only — past session\_answers keep the historical record of what was actually asked at the time.

## 15.3 Removing

- Caregiver taps Delete on a card, confirms once (to prevent accidental removal).
  - Recommended: soft delete (`is_active = false`) rather than hard delete, so historical reports for past sessions still display correctly.
  - A removed card immediately stops being eligible for any new session draw — the elder will never see it again unless the caregiver restores it.
  - Optional "Trash" view lets a caregiver restore a recently removed card within, say, 30 days, before it's hard-deleted for good.
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# 16. Notifications & Alerts

- Weekly email digest to caregiver with the summary described in Section 14.
  - Optional "no session in N days" reminder to the caregiver, useful if the elder needs a nudge.
  - Optional "your library only has N cards" nudge, encouraging the caregiver to add more variety once the pool gets small (e.g. under 10 active cards).
  - Optional streak-milestone note ("3 days in a row!") to the caregiver, shareable with the elder verbally.
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# 17. Security & Privacy

- Elder profiles are not independently authenticated accounts — they are scoped records under a caregiver's login, reducing password-management burden on the elder.
  - Optional elder-facing PIN (4 digits) only to prevent accidental profile-switching on a shared device, not intended as a security boundary.
  - All image storage access via signed, time-limited URLs; never expose raw bucket paths.
  - Uploaded photos may include identifiable people (family photos) — store and transmit over encrypted connections (HTTPS/TLS) and restrict access to the owning caregiver account only.
  - Since the caregiver is the sole content author, no content-moderation pipeline is required — access control on the upload/edit/delete endpoints (only the owning caregiver can touch their elders' cards) is the main safeguard.
  - Provide a clear data-deletion path: caregiver can delete an elder profile and all associated cards/history.
-

# 18. Non-Functional Requirements

| Requirement          | Target                                                                                                                                  |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Elder-view load time | Under 2 seconds to first interactive screen on a tablet/mid-range device.                                                               |
| Card save latency    | Adding or editing a card should feel instant — well under 1 second after upload completes, since there's no generation/processing step. |
| Offline tolerance    | Elder view should gracefully handle brief connectivity drops (retry, don't lose session progress).                                      |
| Browser support      | Latest 2 versions of Chrome, Safari, Edge; iPadOS Safari as a primary target.                                                           |
| Uptime target        | 99.5% for MVP; not medical-grade critical infrastructure.                                                                               |

## 19. Screen-by-Screen Spec

### Elder Mode

| Screen          | Key Elements                                                                                      |
|-----------------|---------------------------------------------------------------------------------------------------|
| Profile Select  | Large tappable avatar/name tiles, one per elder profile on a shared device.                       |
| Home / Start    | Elder's name, a single big "Start" button, optional streak indicator (friendly, not numeric).     |
| Image Reveal    | Full-bleed photo, countdown ring, no other UI chrome.                                             |
| Question        | Large question text (+ audio icon), 2–4 big answer buttons — exactly as the caregiver wrote them. |
| Feedback        | Brief color/animation cue + optional audio, auto-advances after ~2 seconds.                       |
| Session Summary | Warm qualitative summary (stars/smiley), "Play Again" and "Done for now" buttons.                 |

### Caregiver Mode

| Screen           | Key Elements                                                                                                                 |
|------------------|------------------------------------------------------------------------------------------------------------------------------|
| Login / Signup   | Standard auth form + OAuth option.                                                                                           |
| Elder List       | Cards per elder profile with a quick-glance recent accuracy indicator.                                                       |
| Elder Dashboard  | Accuracy trend chart, response-time chart, streak calendar, per-card performance breakdown.                                  |
| Content Library  | Grid of all cards (photo thumbnail + question preview) for an elder, with Edit and Delete on each, and an "Add Card" button. |
| Add / Edit Card  | Single form: photo upload/replace, question prompt, correct answer, wrong-answer choices, Save.                              |
| Session Settings | Difficulty (viewing duration + choice count), round count per session.                                                       |
| Reports          | Weekly digest history, export-to-PDF/CSV button.                                                                             |

## 20. Phased Development Roadmap

### Phase 1 — Core Playable Loop (MVP)

- Caregiver auth + elder profile CRUD.
- Add Card form (photo upload + manual question/choices) and Content Library view.
- Elder Mode: image reveal → question → feedback → session summary, drawing randomly from active cards.
- Basic session logging (answers, correctness, response time) to the database.

### Phase 2 — Content Management Polish

- Edit-card flow (pre-filled form).
- Soft-delete + optional Trash/restore view.

- Per-card performance breakdown feeding into the Library view (e.g. flag cards the elder often misses).

### Phase 3 — Caregiver Insights

- Dashboard charts (accuracy/response-time trends, streaks).
- Weekly email digest, low-library-size nudges.
- PDF/CSV report export.

### Phase 4 — Polish

- Text-to-speech for instructions/questions.
  - Multi-elder / facility support, richer notifications.
  - Bulk card import (e.g. upload several photos at once, then fill in questions in a streamlined batch view).
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## 21. Testing Strategy

- **Unit tests:** scoring logic, difficulty-driven presentation settings, random card sampling from active-only cards.
  - **Integration tests:** add card → appears in session pool → edit card → change reflected → delete card → excluded from pool, end to end.
  - **Usability testing with actual elder users:** observe for hesitation, mis-taps, confusion points — this matters more than most automated testing for this product.
  - **Usability testing with caregivers:** time how long it takes to add a card; anything over ~30–45 seconds per card is worth streamlining.
  - **Accessibility audit:** contrast ratios, tap-target sizes, screen-reader/basic TTS pass.
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## 22. Future Enhancements

- Voice-only mode for elders with visual impairment (audio-described images + spoken answers).
- Multiplayer/social mode (family members submit photos as a weekly "quiz" for the elder).
- Richer cognitive-exercise variety beyond image recall (sequencing, matching, simple word games).
- Clinician-facing export tailored to standard cognitive-screening report formats.
- On-device/offline-first mode for locations with unreliable internet.
- Optional AI-assisted question suggestions as an opt-in convenience feature later, while keeping full manual authoring as the default.